SageMath: Demystifying Graphs

In this series on SageMath, we continue to explore graphs. The focus in this second article is on undirected simple graphs.

In the first article in this series on SageMath, we discussed the different ways to install and use SageMath in our systems. We also covered a few simple examples for SageMath. Please remember that SageMath uses Python 3 syntax. Except for one short snippet of code, we dealt with verbatim Python code throughout. However, by the end of the article, we had used just two lines of pure SageMath code, demonstrating why SageMath has a competitive edge over Python in scientific computing. Those two lines enabled us to draw a particular type of graph called an icosahedral graph. We concluded our discussion with the promise to continue exploring graphs in the upcoming articles in this series on SageMath. Let us begin by doing precisely that.

Over the years, Wikipedia has become an independent tutorial for many topics on science and technology. So, please refer to the Wikipedia article on graph theory for a formal treatment of the subject. If you want to learn graph theory in more detail, I recommend a textbook titled 'Introduction to Graph Theory' by Douglas B. West. However, my treatment of graph theory will be pretty informal.

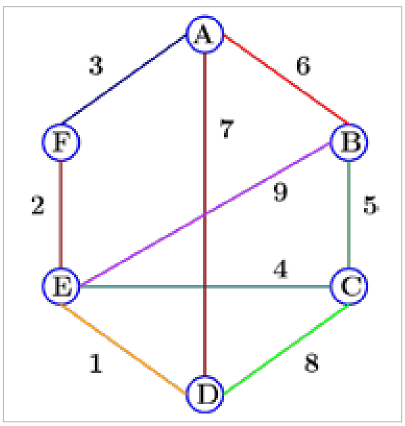

Figure 1: The undirected simple graph *Graph_1*

Consider a practical example to understand how graphs help us model real-world problems. Picture a group of six people: Amrita, Biju, Chitra, Dev, Ekta, and Farhan sharing a table at a function. Some of these people are friends, while others are unknown to each other. We can represent their familiarity with each other using a graph. Consider the graph shown in Figure 1. We call this *Graph_1* for ease of reference during explanations.

The circles in the graph are called vertices, nodes, or points. They represent the objects under discussion, in this case, the six people attending the party. The six people, Amrita, Biju, Chitra, Dev, Ekta, and Farhan, are represented by vertices labelled with upper case letters A to F, respectively. The lines connecting the vertices are called edges. In this case, the edges are used to denote whether two people are friends or not. For example, from Figure 1, it can be observed that Dev is friends with Amrita, Chitra, and Ekta but not with Biju and Farhan. The edges of a graph can also have labels or numerical weights. In Figure 1, the edge weights denote the number of years two people have been friends. For example, Dev and Chitra have been friends for eight years, whereas Dev and Ekta have been friends for just one year.

Now, let us delve into the process of creating this graph (*Graph_1*) using SageMath. Take a look at the SageMath code, saved as *graph1.sagews*, provided below (line numbers have been